VEXOVAULT GOLDEN TEST FIXTURE: CHAPTER 1 - ELECTROSTATICS

Known String Page 1: VEXOVAULT_VERIFY_TOKEN_ALPHA_001

Coulomb's Law states that the electrical force between two charged objects is directly proportional to the product of the qua

Formula: F = (1 / (4 * pi * epsilon_0)) * (q1 * q2) / r^2

SI unit of permittivity is C^2 N^-1 m^-2. The value of epsilon_0 is 8.854 x 10^-12 F/m.

Known String Page 2: VEXOVAULT_VERIFY_TOKEN_BETA_002

Unicode math symbols: x² + y² = r²

Subscripts in chemistry: H·O + CO· · H·CO·

Scientific notation: 10·· C and 1.6 × 10·1· C

Operators: a · b, x · y, a · b, · + · = ·

Known String Page 3: VEXOVAULT_VERIFY_TOKEN_GAMMA_003

Table 1.1: Physical Constants and Units
Quantity	Symbol	Value	SI Unit
Permittivity of free space	epsilon_0	8.854 x 10^-12	C^2 N^-1 m^-2
Elementary charge	e	1.602 x 10^-19	C
Mass of electron	m_e	9.109 x 10^-31	kg
Speed of light	c	2.998 x 10^8	m/s

Known String Page 4: VEXOVAULT_VERIFY_TOKEN_DELTA_004

Column A: Electric Field	Column B: Electric Potential
The electric field E is defined as the force per unit charge exerted on a test positive charge placed at that point in space.	The electric potential V at any point in space is the work done per unit charge by an external agent in moving a test charge from infinity to that point without acceleration.

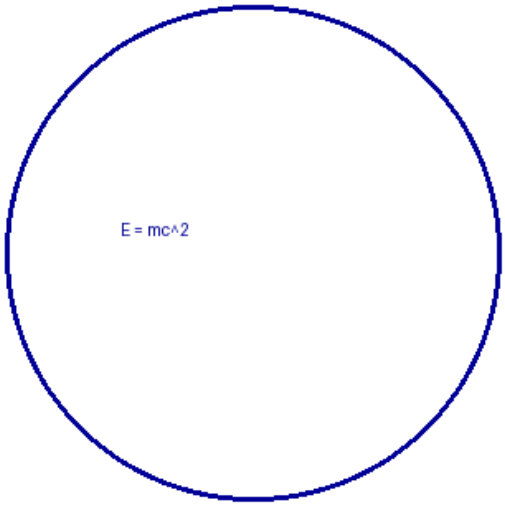
Known String Page 5: VEXOVAULT_VERIFY_TOKEN_EPSILON_005

Worked Derivation: Electric Field of an Electric Dipole on the Axial Line

Given:
An electric dipole consisting of two charges -q and +q separated by a distance 2a.
Step 1: The electric field E1 at point P due to charge +q at distance (r - a) is E1 = q / (4 * pi * epsilon_0 * (r - a)^2) along BP.
Step 2: The electric field E2 at point P due to charge -q at distance (r + a) is E2 = q / (4 * pi * epsilon_0 * (r + a)^2) along PA.
Step 3: Since E1 and E2 act along the same line in opposite directions, net electric field E = E1 - E2.
Step 4: E = (q / (4 * pi * epsilon_0)) * [1/(r - a)^2 - 1/(r + a)^2] = (q / (4 * pi * epsilon_0)) * [4ar / (r^2 - a^2)^2].
Step 5: For a short dipole where r >> a, we have E_axial = (2 * p) / (4 * pi * epsilon_0 * r^3), where p = 2aq is the dipole moment.
Result:
The electric field varies inversely with the cube of the distance from the dipole center.

SCANNED OCR TEST PAGE - OPTICAL CHARACTER RECOGNITION

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Chapter 1: Synthetic Chapter Section - Page 8

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